

2.3 Functions

2.3.1 Introduction

In many instances we assign to each element of a set a particular element of a second set (which may be the same as the first). For example, suppose that each student in a discrete mathematics class is assigned a letter grade from the set $\{A, B, C, D, F\}$. And suppose that the grades are A for Adams, C for Chou, B for Goodfriend, A for Rodriguez, and F for Stevens. This assignment of grades is illustrated in Figure 1.

This assignment is an example of a function. The concept of a function is extremely important in mathematics and computer science. For example, in discrete mathematics functions are used in the definition of such discrete structures as sequences and strings. Functions are also used to represent how long it takes a computer to solve problems of a given size. Many computer programs and subroutines are designed to calculate values of functions. Recursive functions, which are functions defined in terms of themselves, are used throughout computer science; they will be studied in Chapter 5. This section reviews the basic concepts involving functions needed in discrete mathematics.

Definition 1

Let A and B be nonempty sets. A *function* f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f : A \rightarrow B$.

Remark: Functions are sometimes also called **mappings** or **transformations**.

Assessment

Functions are specified in many different ways. Sometimes we explicitly state the assignments, as in Figure 1. Often we give a formula, such as $f(x) = x + 1$, to define a function. Other times we use a computer program to specify a function.

A function $f : A \rightarrow B$ can also be defined in terms of a relation from A to B . Recall from Section 2.1 that a relation from A to B is just a subset of $A \times B$. A relation from A to B that contains one, and only one, ordered pair (a, b) for every element $a \in A$, defines a function f from A to B . This function is defined by the assignment $f(a) = b$, where (a, b) is the unique ordered pair in the relation that has a as its first element.

Definition 2

If f is a function from A to B , we say that A is the *domain* of f and B is the *codomain* of f . If $f(a) = b$, we say that b is the *image* of a and a is a *preimage* of b . The *range*, or *image*, of f is the set of all images of elements of A . Also, if f is a function from A to B , we say that f *maps* A to B .

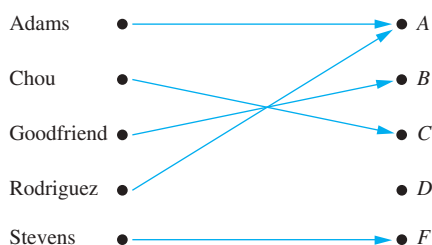


FIGURE 1 Assignment of grades in a discrete mathematics class.

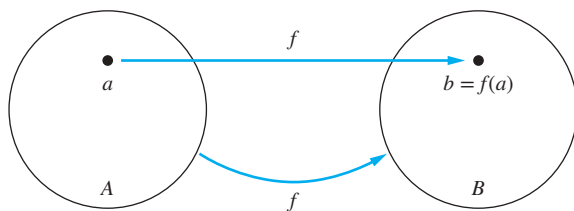


FIGURE 2 The function f maps A to B .

Figure 2 represents a function f from A to B .

Remark: Note that the codomain of a function from A to B is the set of all possible values of such a function (that is, all elements of B), and the range is the set of all values of $f(a)$ for $a \in A$, and is always a subset of the codomain. That is, the codomain is the set of possible values of the function and the range is the set of all elements of the codomain that are achieved as the value of f for at least one element of the domain.

When we define a function we specify its domain, its codomain, and the mapping of elements of the domain to elements in the codomain. Two functions are **equal** when they have the same domain, have the same codomain, and map each element of their common domain to the same element in their common codomain. Note that if we change either the domain or the codomain of a function, then we obtain a different function. If we change the mapping of elements, then we also obtain a different function.

Examples 1–5 provide examples of functions. In each case, we describe the domain, the codomain, the range, and the assignment of values to elements of the domain.

EXAMPLE 1 What are the domain, codomain, and range of the function that assigns grades to students described in the first paragraph of the introduction of this section?

Solution: Let G be the function that assigns a grade to a student in our discrete mathematics class. Note that $G(\text{Adams}) = A$, for instance. The domain of G is the set $\{\text{Adams, Chou, Goodfriend, Rodriguez, Stevens}\}$, and the codomain is the set $\{A, B, C, D, F\}$. The range of G is the set $\{A, B, C, F\}$, because each grade except D is assigned to some student. ◀

EXAMPLE 2 Let R be the relation with ordered pairs $(\text{Abdul}, 22)$, $(\text{Brenda}, 24)$, $(\text{Carla}, 21)$, $(\text{Desire}, 22)$, $(\text{Eddie}, 24)$, and $(\text{Felicia}, 22)$. Here each pair consists of a graduate student and this student's age. Specify a function determined by this relation.

Solution: If f is a function specified by R , then $f(\text{Abdul}) = 22$, $f(\text{Brenda}) = 24$, $f(\text{Carla}) = 21$, $f(\text{Desire}) = 22$, $f(\text{Eddie}) = 24$, and $f(\text{Felicia}) = 22$. [Here, $f(x)$ is the age of x , where x is a student.] For the domain, we take the set $\{\text{Abdul, Brenda, Carla, Desire, Eddie, Felicia}\}$. We also need to specify a codomain, which needs to contain all possible ages of students. Because it is highly likely that all students are less than 100 years old, we can take the set of positive integers less than 100 as the codomain. (Note that we could choose a different codomain, such as the set of all positive integers or the set of positive integers between 10 and 90, but that would change the function. Using this codomain will also allow us to extend the function by adding the names and ages of more students later.) The range of the function we have specified is the set of different ages of these students, which is the set $\{21, 22, 24\}$. ◀

EXAMPLE 3 Let f be the function that assigns the last two bits of a bit string of length 2 or greater to that string. For example, $f(11010) = 10$. Then, the domain of f is the set of all bit strings of length 2 or greater, and both the codomain and range are the set $\{00, 01, 10, 11\}$. ◀

Extra Examples ▶

EXAMPLE 4 Let $f: \mathbf{Z} \rightarrow \mathbf{Z}$ assign the square of an integer to this integer. Then, $f(x) = x^2$, where the domain of f is the set of all integers, the codomain of f is the set of all integers, and the range of f is the set of all integers that are perfect squares, namely, $\{0, 1, 4, 9, \dots\}$. 

EXAMPLE 5 The domain and codomain of functions are often specified in programming languages. For instance, the Java statement

```
int floor(float real){ ... }
```

and the C++ function statement

```
int function (float x){ ... }
```

both tell us that the domain of the floor function is the set of real numbers (represented by floating point numbers) and its codomain is the set of integers. 

A function is called **real-valued** if its codomain is the set of real numbers, and it is called **integer-valued** if its codomain is the set of integers. Two real-valued functions or two integer-valued functions with the same domain can be added, as well as multiplied.

Definition 3

Let f_1 and f_2 be functions from A to $\mathbf{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined for all $x \in A$ by

$$\begin{aligned}(f_1 + f_2)(x) &= f_1(x) + f_2(x), \\ (f_1 f_2)(x) &= f_1(x) f_2(x).\end{aligned}$$

Note that the functions $f_1 + f_2$ and $f_1 f_2$ have been defined by specifying their values at x in terms of the values of f_1 and f_2 at x .

EXAMPLE 6 Let f_1 and f_2 be functions from $\mathbf{R}$ to $\mathbf{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution: From the definition of the sum and product of functions, it follows that

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

and

$$(f_1 f_2)(x) = x^2(x - x^2) = x^3 - x^4. \quad \text{◀}$$

When f is a function from A to B , the image of a subset of A can also be defined.

Definition 4

Let f be a function from A to B and let S be a subset of A . The *image* of S under the function f is the subset of B that consists of the images of the elements of S . We denote the image of S by $f(S)$, so

$$f(S) = \{t \mid \exists s \in S (t = f(s))\}.$$

We also use the shorthand $\{f(s) \mid s \in S\}$ to denote this set.

Remark: The notation $f(S)$ for the image of the set S under the function f is potentially ambiguous. Here, $f(S)$ denotes a set, and not the value of the function f for the set S .

EXAMPLE 7 Let $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4\}$ with $f(a) = 2, f(b) = 1, f(c) = 4, f(d) = 1$, and $f(e) = 1$. The image of the subset $S = \{b, c, d\}$ is the set $f(S) = \{1, 4\}$. ◀

2.3.2 One-to-One and Onto Functions

Some functions never assign the same value to two different domain elements. These functions are said to be **one-to-one**.

Definition 5

A function f is said to be *one-to-one*, or an *injection*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be *injective* if it is one-to-one.

Note that a function f is one-to-one if and only if $f(a) \neq f(b)$ whenever $a \neq b$. This way of expressing that f is one-to-one is obtained by taking the contrapositive of the implication in the definition.

Remark: We can express that f is one-to-one using quantifiers as $\forall a \forall b (f(a) = f(b) \rightarrow a = b)$ or equivalently $\forall a \forall b (a \neq b \rightarrow f(a) \neq f(b))$, where the universe of discourse is the domain of the function.

Assessment

We illustrate this concept by giving examples of functions that are one-to-one and other functions that are not one-to-one.

EXAMPLE 8

Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4, f(b) = 5, f(c) = 1$, and $f(d) = 3$ is one-to-one.

Extra Examples

Solution: The function f is one-to-one because f takes on different values at the four elements of its domain. This is illustrated in Figure 3. ◀

EXAMPLE 9

Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

Solution: The function $f(x) = x^2$ is not one-to-one because, for instance, $f(1) = f(-1) = 1$, but $1 \neq -1$. ◀

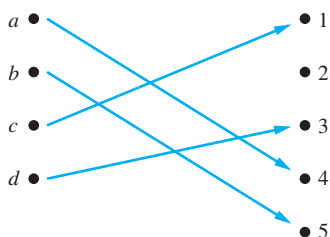


FIGURE 3 A one-to-one function.

Remark: The function $f(x) = x^2$ with domain $\mathbf{Z}^+$ is one-to-one. (See the explanation in Example 12 to see why.) This is a different function from the function in Example 9 because of the difference in their domains.

EXAMPLE 10 Determine whether the function $f(x) = x + 1$ from the set of real numbers to itself is one-to-one.

Solution: Suppose that x and y are real numbers with $f(x) = f(y)$, so that $x + 1 = y + 1$. This means that $x = y$. Hence, $f(x) = x + 1$ is a one-to-one function from $\mathbf{R}$ to $\mathbf{R}$. ◀

EXAMPLE 11 Suppose that each worker in a group of employees is assigned a job from a set of possible jobs, each to be done by a single worker. In this situation, the function f that assigns a job to each worker is one-to-one. To see this, note that if x and y are two different workers, then $f(x) \neq f(y)$ because the two workers x and y must be assigned different jobs. ◀

We now give some conditions that guarantee that a function is one-to-one.

Definition 6

A function f whose domain and codomain are subsets of the set of real numbers is called *increasing* if $f(x) \leq f(y)$, and *strictly increasing* if $f(x) < f(y)$, whenever $x < y$ and x and y are in the domain of f . Similarly, f is called *decreasing* if $f(x) \geq f(y)$, and *strictly decreasing* if $f(x) > f(y)$, whenever $x < y$ and x and y are in the domain of f . (The word *strictly* in this definition indicates a strict inequality.)

Remark: A function f is increasing if $\forall x \forall y (x < y \rightarrow f(x) \leq f(y))$, strictly increasing if $\forall x \forall y (x < y \rightarrow f(x) < f(y))$, decreasing if $\forall x \forall y (x < y \rightarrow f(x) \geq f(y))$, and strictly decreasing if $\forall x \forall y (x < y \rightarrow f(x) > f(y))$, where the universe of discourse is the domain of f .

EXAMPLE 12 The function $f(x) = x^2$ from $\mathbf{R}^+$ to $\mathbf{R}^+$ is strictly increasing. To see this, suppose that x and y are positive real numbers with $x < y$. Multiplying both sides of this inequality by x gives $x^2 < xy$. Similarly, multiplying both sides by y gives $xy < y^2$. Hence, $f(x) = x^2 < xy < y^2 = f(y)$. However, the function $f(x) = x^2$ from $\mathbf{R}$ to the set of nonnegative real numbers is not strictly increasing because $-1 < 0$, but $f(-1) = (-1)^2 = 1$ is not less than $f(0) = 0^2 = 0$. ◀

From these definitions, it can be shown (see Exercises 26 and 27) that a function that is either strictly increasing or strictly decreasing must be one-to-one. However, a function that is increasing, but not strictly increasing, or decreasing, but not strictly decreasing, is not one-to-one.

For some functions the range and the codomain are equal. That is, every member of the codomain is the image of some element of the domain. Functions with this property are called **onto** functions.

Definition 7

A function f from A to B is called *onto*, or a *surjection*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called *surjective* if it is onto.

Remark: A function f is onto if $\forall y \exists x (f(x) = y)$, where the domain for x is the domain of the function and the domain for y is the codomain of the function.

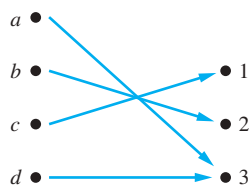


FIGURE 4 An onto function.

We now give examples of onto functions and functions that are not onto.

EXAMPLE 13 Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function?

Extra Examples ➤

Solution: Because all three elements of the codomain are images of elements in the domain, we see that f is onto. This is illustrated in Figure 4. Note that if the codomain were $\{1, 2, 3, 4\}$, then f would not be onto. ◀

EXAMPLE 14 Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

Solution: The function f is not onto because there is no integer x with $x^2 = -1$, for instance. ◀

EXAMPLE 15 Is the function $f(x) = x + 1$ from the set of integers to the set of integers onto?

Solution: This function is onto, because for every integer y there is an integer x such that $f(x) = y$. To see this, note that $f(x) = y$ if and only if $x + 1 = y$, which holds if and only if $x = y - 1$. (Note that $y - 1$ is also an integer, and so, is in the domain of f .) ◀

EXAMPLE 16 Consider the function f in Example 11 that assigns jobs to workers. The function f is onto if for every job there is a worker assigned this job. The function f is not onto when there is at least one job that has no worker assigned it. ◀

Definition 8

The function f is a *one-to-one correspondence*, or a *bijection*, if it is both one-to-one and onto. We also say that such a function is *bijection*.

Examples 16 and 17 illustrate the concept of a bijection.

EXAMPLE 17 Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f a bijection?

Solution: The function f is one-to-one and onto. It is one-to-one because no two values in the domain are assigned the same function value. It is onto because all four elements of the codomain are images of elements in the domain. Hence, f is a bijection. ◀

Figure 5 displays four functions where the first is one-to-one but not onto, the second is onto but not one-to-one, the third is both one-to-one and onto, and the fourth is neither one-to-one nor onto. The fifth correspondence in Figure 5 is not a function, because it sends an element to two different elements.

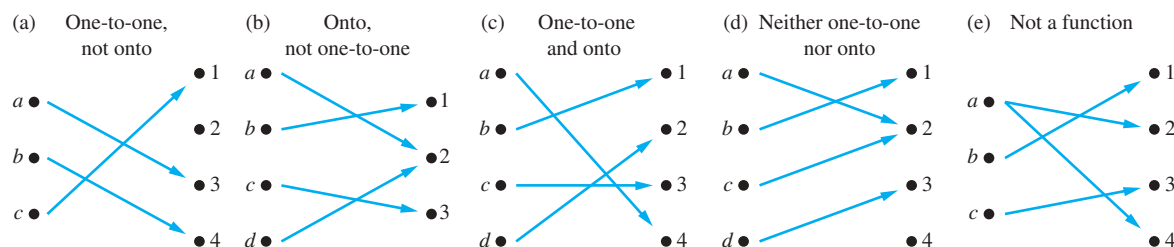


FIGURE 5 Examples of different types of correspondences.

Suppose that f is a function from a set A to itself. If A is finite, then f is one-to-one if and only if it is onto. (This follows from the result in Exercise 74.) This is not necessarily the case if A is infinite (as will be shown in Section 2.5).

EXAMPLE 18 Let A be a set. The *identity function* on A is the function $\iota_A : A \rightarrow A$, where

$$\iota_A(x) = x$$

for all $x \in A$. In other words, the identity function ι_A is the function that assigns each element to itself. The function ι_A is one-to-one and onto, so it is a bijection. (Note that ι is the Greek letter iota.)

For future reference, we summarize what needs to be shown to establish whether a function is one-to-one and whether it is onto. It is instructive to review Examples 8–17 in light of this summary.

Suppose that $f : A \rightarrow B$.

To show that f is injective Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

To show that f is not injective Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

To show that f is surjective Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

To show that f is not surjective Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

2.3.3 Inverse Functions and Compositions of Functions

Now consider a one-to-one correspondence f from the set A to the set B . Because f is an onto function, every element of B is the image of some element in A . Furthermore, because f is also a one-to-one function, every element of B is the image of a *unique* element of A . Consequently, we can define a new function from B to A that reverses the correspondence given by f . This leads to Definition 9.

Definition 9

Let f be a one-to-one correspondence from the set A to the set B . The *inverse function* of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.

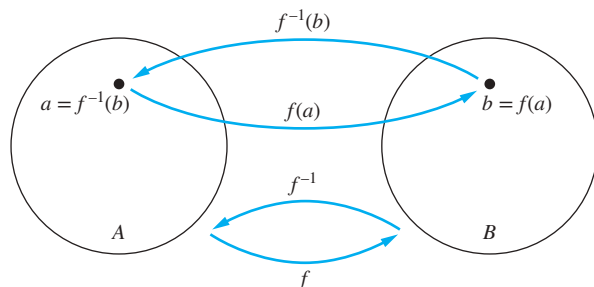


FIGURE 6 The function f^{-1} is the inverse of function f .

Remark: Be sure not to confuse the function f^{-1} with the function $1/f$, which is the function that assigns to each x in the domain the value $1/f(x)$. Notice that the latter makes sense only when $f(x)$ is a nonzero real number.

Figure 6 illustrates the concept of an inverse function.

If a function f is not a one-to-one correspondence, we cannot define an inverse function of f . When f is not a one-to-one correspondence, either it is not one-to-one or it is not onto. If f is not one-to-one, some element b in the codomain is the image of more than one element in the domain. If f is not onto, for some element b in the codomain, no element a in the domain exists for which $f(a) = b$. Consequently, if f is not a one-to-one correspondence, we cannot assign to each element b in the codomain a unique element a in the domain such that $f(a) = b$ (because for some b there is either more than one such a or no such a).

A one-to-one correspondence is called **invertible** because we can define an inverse of this function. A function is **not invertible** if it is not a one-to-one correspondence, because the inverse of such a function does not exist.

EXAMPLE 19 Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$. ◀

EXAMPLE 20 Let $f : \mathbf{Z} \rightarrow \mathbf{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f has an inverse because it is a one-to-one correspondence, as follows from Examples 10 and 15. To reverse the correspondence, suppose that y is the image of x , so that $y = x + 1$. Then $x = y - 1$. This means that $y - 1$ is the unique element of $\mathbf{Z}$ that is sent to y by f . Consequently, $f^{-1}(y) = y - 1$. ◀

EXAMPLE 21 Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ with $f(x) = x^2$. Is f invertible?

Solution: Because $f(-2) = f(2) = 4$, f is not one-to-one. If an inverse function were defined, it would have to assign two elements to 4. Hence, f is not invertible. (Note we can also show that f is not invertible because it is not onto.) ◀

Sometimes we can restrict the domain or the codomain of a function, or both, to obtain an invertible function, as Example 22 illustrates.

EXAMPLE 22 Show that if we restrict the function $f(x) = x^2$ in Example 21 to a function from the set of all nonnegative real numbers to the set of all nonnegative real numbers, then f is invertible.

Solution: The function $f(x) = x^2$ from the set of nonnegative real numbers to the set of nonnegative real numbers is one-to-one. To see this, note that if $f(x) = f(y)$, then $x^2 = y^2$, so $x^2 - y^2 = (x + y)(x - y) = 0$. This means that $x + y = 0$ or $x - y = 0$, so $x = -y$ or $x = y$. Because both x and y are nonnegative, we must have $x = y$. So, this function is one-to-one. Furthermore, $f(x) = x^2$ is onto when the codomain is the set of all nonnegative real numbers, because each nonnegative real number has a square root. That is, if y is a nonnegative real number, there exists a nonnegative real number x such that $x = \sqrt{y}$, which means that $x^2 = y$. Because the function $f(x) = x^2$ from the set of nonnegative real numbers to the set of nonnegative real numbers is one-to-one and onto, it is invertible. Its inverse is given by the rule $f^{-1}(y) = \sqrt{y}$. ◀

Definition 10

Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The *composition* of the functions f and g , denoted for all $a \in A$ by $f \circ g$, is the function from A to C defined by

$$(f \circ g)(a) = f(g(a)).$$

In other words, $f \circ g$ is the function that assigns to the element a of A the element assigned by f to $g(a)$. The domain of $f \circ g$ is the domain of g . The range of $f \circ g$ is the image of the range of g with respect to the function f . That is, to find $(f \circ g)(a)$ we first apply the function g to a to obtain $g(a)$ and then we apply the function f to the result $g(a)$ to obtain $(f \circ g)(a) = f(g(a))$. Note that the composition $f \circ g$ cannot be defined unless the range of g is a subset of the domain of f . In Figure 7 the composition of functions is shown.

EXAMPLE 23 Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g , and what is the composition of g and f ?

Solution: The composition $f \circ g$ is defined by $(f \circ g)(a) = f(g(a)) = f(b) = 2$, $(f \circ g)(b) = f(g(b)) = f(c) = 1$, and $(f \circ g)(c) = f(g(c)) = f(a) = 3$.

Note that $g \circ f$ is not defined, because the range of f is not a subset of the domain of g . ◀

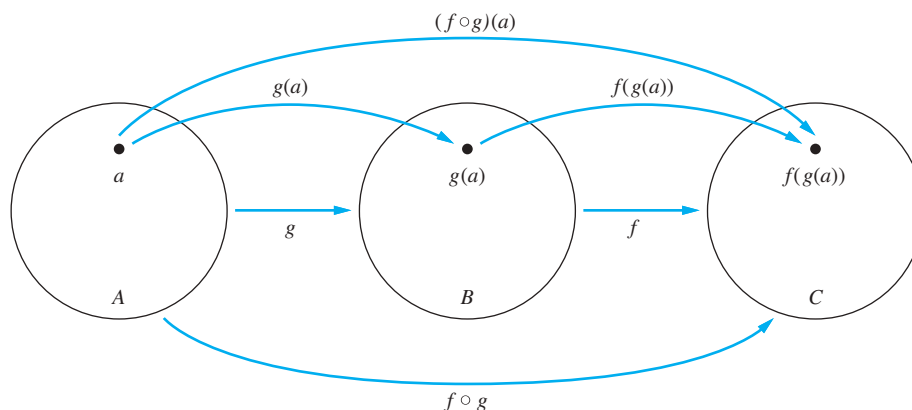


FIGURE 7 The composition of the functions f and g .

EXAMPLE 24 Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

Solution: Both the compositions $f \circ g$ and $g \circ f$ are defined. Moreover,

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

and

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11. \quad \blacktriangleleft$$

Remark: Note that even though $f \circ g$ and $g \circ f$ are defined for the functions f and g in Example 24, $f \circ g$ and $g \circ f$ are not equal. In other words, the commutative law does not hold for the composition of functions.

EXAMPLE 25 Let f and g be the functions defined by $f : \mathbf{R} \rightarrow \mathbf{R}^+ \cup \{0\}$ with $f(x) = x^2$ and $g : \mathbf{R}^+ \cup \{0\} \rightarrow \mathbf{R}$ with $g(x) = \sqrt{x}$ (where $\sqrt{x}$ is the nonnegative square root of x). What is the function $(f \circ g)(x)$?

Solution: The domain of $(f \circ g)(x) = f(g(x))$ is the domain of g , which is $\mathbf{R}^+ \cup \{0\}$, the set of nonnegative real numbers. If x is a nonnegative real number, we have $(f \circ g)(x) = f(g(x)) = f(\sqrt{x}) = (\sqrt{x})^2 = x$. The range of $f \circ g$ is the image of the range of g with respect to the function f . This is the set $\mathbf{R}^+ \cup \{0\}$, the set of nonnegative real numbers. Summarizing, $f : \mathbf{R}^+ \cup \{0\} \rightarrow \mathbf{R}^+ \cup \{0\}$ and $f(g(x)) = x$ for all x . $\blacktriangleleft$

When the composition of a function and its inverse is formed, in either order, an identity function is obtained. To see this, suppose that f is a one-to-one correspondence from the set A to the set B . Then the inverse function f^{-1} exists and is a one-to-one correspondence from B to A . The inverse function reverses the correspondence of the original function, so $f^{-1}(b) = a$ when $f(a) = b$, and $f(a) = b$ when $f^{-1}(b) = a$. Hence,

$$(f^{-1} \circ f)(a) = f^{-1}(f(a)) = f^{-1}(b) = a,$$

and

$$(f \circ f^{-1})(b) = f(f^{-1}(b)) = f(a) = b.$$

Consequently $f^{-1} \circ f = \iota_A$ and $f \circ f^{-1} = \iota_B$, where ι_A and ι_B are the identity functions on the sets A and B , respectively. That is, $(f^{-1})^{-1} = f$.

2.3.4 The Graphs of Functions

We can associate a set of pairs in $A \times B$ to each function from A to B . This set of pairs is called the **graph** of the function and is often displayed pictorially to aid in understanding the behavior of the function.

Definition 11

Let f be a function from the set A to the set B . The *graph* of the function f is the set of ordered pairs $\{(a, b) \mid a \in A \text{ and } f(a) = b\}$.

From the definition, the graph of a function f from A to B is the subset of $A \times B$ containing the ordered pairs with the second entry equal to the element of B assigned by f to the first entry. Also, note that the graph of a function f from A to B is the same as the relation from A to B determined by the function f , as described on Section 2.3.1.

EXAMPLE 26 Display the graph of the function $f(n) = 2n + 1$ from the set of integers to the set of integers.

Solution: The graph of f is the set of ordered pairs of the form $(n, 2n + 1)$, where n is an integer. This graph is displayed in Figure 8. ◀

EXAMPLE 27 Display the graph of the function $f(x) = x^2$ from the set of integers to the set of integers.

Solution: The graph of f is the set of ordered pairs of the form $(x, f(x)) = (x, x^2)$, where x is an integer. This graph is displayed in Figure 9. ◀

2.3.5 Some Important Functions

Next, we introduce two important functions in discrete mathematics, namely, the floor and ceiling functions. Let x be a real number. The floor function rounds x down to the closest integer less than or equal to x , and the ceiling function rounds x up to the closest integer greater than or equal to x . These functions are often used when objects are counted. They play an important role in the analysis of the number of steps used by procedures to solve problems of a particular size.

Definition 12

The *floor function* assigns to the real number x the largest integer that is less than or equal to x . The value of the floor function at x is denoted by $\lfloor x \rfloor$. The *ceiling function* assigns to the real number x the smallest integer that is greater than or equal to x . The value of the ceiling function at x is denoted by $\lceil x \rceil$.

Remark: The floor function is often also called the *greatest integer function*. It is often denoted by $\lfloor x \rfloor$.

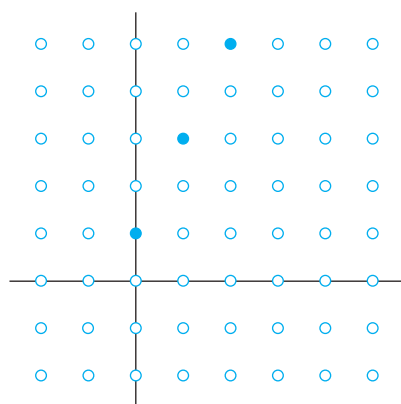


FIGURE 8 The graph of $f(n) = 2n + 1$ from $\mathbb{Z}$ to $\mathbb{Z}$.

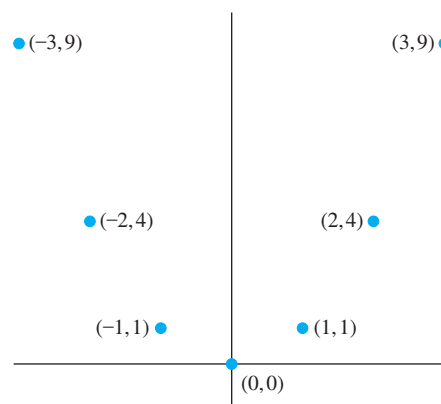


FIGURE 9 The graph of $f(x) = x^2$ from $\mathbb{Z}$ to $\mathbb{Z}$.

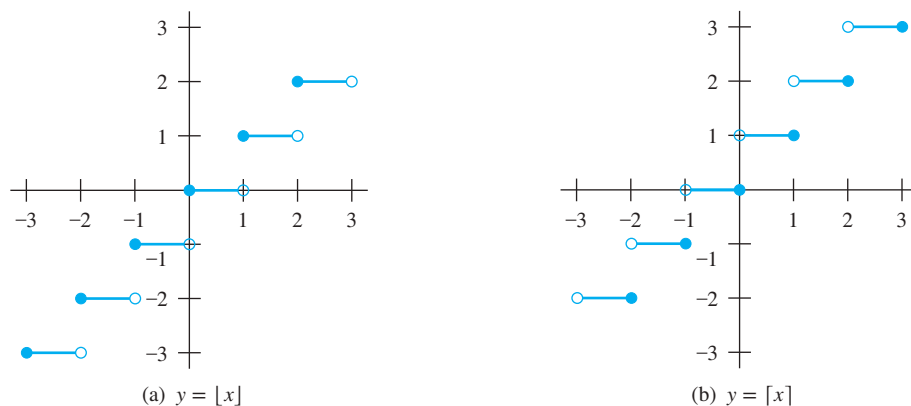


FIGURE 10 Graphs of the (a) floor and (b) ceiling functions.

EXAMPLE 28 These are some values of the floor and ceiling functions:

$$\lfloor \frac{1}{2} \rfloor = 0, \lceil \frac{1}{2} \rceil = 1, \lfloor -\frac{1}{2} \rfloor = -1, \lceil -\frac{1}{2} \rceil = 0, \lfloor 3.1 \rfloor = 3, \lceil 3.1 \rceil = 4, \lfloor 7 \rfloor = 7, \lceil 7 \rceil = 7. \quad \blacktriangleleft$$

We display the graphs of the floor and ceiling functions in Figure 10. In Figure 10(a) we display the graph of the floor function $\lfloor x \rfloor$. Note that this function has the same value throughout the interval $[n, n+1)$, namely n , and then it jumps up to $n+1$ when $x = n+1$. In Figure 10(b) we display the graph of the ceiling function $\lceil x \rceil$. Note that this function has the same value throughout the interval $(n, n+1]$, namely $n+1$, and then jumps to $n+2$ when x is a little larger than $n+1$.

Links 

The floor and ceiling functions are useful in a wide variety of applications, including those involving data storage and data transmission. Consider Examples 29 and 30, typical of basic calculations done when database and data communications problems are studied.

EXAMPLE 29 Data stored on a computer disk or transmitted over a data network are usually represented as a string of bytes. Each byte is made up of 8 bits. How many bytes are required to encode 100 bits of data?

Solution: To determine the number of bytes needed, we determine the smallest integer that is at least as large as the quotient when 100 is divided by 8, the number of bits in a byte. Consequently, $\lceil 100/8 \rceil = \lceil 12.5 \rceil = 13$ bytes are required. 

EXAMPLE 30 In asynchronous transfer mode (ATM) (a communications protocol used on backbone networks), data are organized into cells of 53 bytes. How many ATM cells can be transmitted in 1 minute over a connection that transmits data at the rate of 500 kilobits per second?

Solution: In 1 minute, this connection can transmit $500,000 \cdot 60 = 30,000,000$ bits. Each ATM cell is 53 bytes long, which means that it is $53 \cdot 8 = 424$ bits long. To determine the number of cells that can be transmitted in 1 minute, we determine the largest integer not exceeding the quotient when 30,000,000 is divided by 424. Consequently, $\lfloor 30,000,000/424 \rfloor = 70,754$ ATM cells can be transmitted in 1 minute over a 500 kilobit per second connection. 

Table 1, with x denoting a real number, displays some simple but important properties of the floor and ceiling functions. Because these functions appear so frequently in discrete

TABLE 1 Useful Properties of the Floor and Ceiling Functions. $(n \text{ is an integer, } x \text{ is a real number})$

(1a) $\lfloor x \rfloor = n$ if and only if $n \leq x < n + 1$

(1b) $\lceil x \rceil = n$ if and only if $n - 1 < x \leq n$

(1c) $\lfloor x \rfloor = n$ if and only if $x - 1 < n \leq x$

(1d) $\lceil x \rceil = n$ if and only if $x \leq n < x + 1$

(2) $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$

(3a) $\lfloor -x \rfloor = -\lceil x \rceil$

(3b) $\lceil -x \rceil = -\lfloor x \rfloor$

(4a) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$

(4b) $\lceil x + n \rceil = \lceil x \rceil + n$

mathematics, it is useful to look over these identities. Each property in this table can be established using the definitions of the floor and ceiling functions. Properties (1a), (1b), (1c), and (1d) follow directly from these definitions. For example, (1a) states that $\lfloor x \rfloor = n$ if and only if the integer n is less than or equal to x and $n + 1$ is larger than x . This is precisely what it means for n to be the greatest integer not exceeding x , which is the definition of $\lfloor x \rfloor = n$. Properties (1b), (1c), and (1d) can be established similarly. We will prove property (4a) using a direct proof.

Proof: Suppose that $\lfloor x \rfloor = m$, where m is a positive integer. By property (1a), it follows that $m \leq x < m + 1$. Adding n to all three quantities in this chain of two inequalities shows that $m + n \leq x + n < m + n + 1$. Using property (1a) again, we see that $\lfloor x + n \rfloor = m + n = \lfloor x \rfloor + n$. This completes the proof. Proofs of the other properties are left as exercises. 

The floor and ceiling functions enjoy many other useful properties besides those displayed in Table 1. There are also many statements about these functions that may appear to be correct, but actually are not. We will consider statements about the floor and ceiling functions in Examples 31 and 32.

A useful approach for considering statements about the floor function is to let $x = n + \epsilon$, where $n = \lfloor x \rfloor$ is an integer, and ϵ , the fractional part of x , satisfies the inequality $0 \leq \epsilon < 1$. Similarly, when considering statements about the ceiling function, it is useful to write $x = n - \epsilon$, where $n = \lceil x \rceil$ is an integer and $0 \leq \epsilon < 1$.

EXAMPLE 31 Prove that if x is a real number, then $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor$.

Extra Examples 

Solution: To prove this statement we let $x = n + \epsilon$, where n is an integer and $0 \leq \epsilon < 1$. There are two cases to consider, depending on whether ϵ is less than, or greater than or equal to $\frac{1}{2}$. (The reason we choose these two cases will be made clear in the proof.)

Links 

We first consider the case when $0 \leq \epsilon < \frac{1}{2}$. In this case, $2x = 2n + 2\epsilon$ and $\lfloor 2x \rfloor = 2n$ because $0 \leq 2\epsilon < 1$. Similarly, $x + \frac{1}{2} = n + (\frac{1}{2} + \epsilon)$, so $\lfloor x + \frac{1}{2} \rfloor = n$, because $0 < \frac{1}{2} + \epsilon < 1$. Consequently, $\lfloor 2x \rfloor = 2n$ and $\lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor = n + n = 2n$.

Next, we consider the case when $\frac{1}{2} \leq \epsilon < 1$. In this case, $2x = 2n + 2\epsilon = (2n + 1) + (2\epsilon - 1)$. Because $0 \leq 2\epsilon - 1 < 1$, it follows that $\lfloor 2x \rfloor = 2n + 1$. Because

$\lfloor x + \frac{1}{2} \rfloor = \lfloor n + (\frac{1}{2} + \epsilon) \rfloor = \lfloor n + 1 + (\epsilon - \frac{1}{2}) \rfloor$ and $0 \leq \epsilon - \frac{1}{2} < 1$, it follows that $\lfloor x + \frac{1}{2} \rfloor = n + 1$. Consequently, $\lfloor 2x \rfloor = 2n + 1$ and $\lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor = n + (n + 1) = 2n + 1$. This concludes the proof. 

EXAMPLE 32 Prove or disprove that $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$ for all real numbers x and y .

Solution: Although this statement may appear reasonable, it is false. A counterexample is supplied by $x = \frac{1}{2}$ and $y = \frac{1}{2}$. With these values we find that $\lfloor x + y \rfloor = \lfloor \frac{1}{2} + \frac{1}{2} \rfloor = \lfloor 1 \rfloor = 1$, but $\lfloor x \rfloor + \lfloor y \rfloor = \lfloor \frac{1}{2} \rfloor + \lfloor \frac{1}{2} \rfloor = 1 + 1 = 2$. 

There are certain types of functions that will be used throughout the text. These include polynomial, logarithmic, and exponential functions. A brief review of the properties of these functions needed in this text is given in Appendix 2. In this book the notation $\log x$ will be used to denote the logarithm to the base 2 of x , because 2 is the base that we will usually use for logarithms. We will denote logarithms to the base b , where b is any real number greater than 1, by $\log_b x$, and the natural logarithm by $\ln x$.

Another function we will use throughout this text is the **factorial function** $f: \mathbf{N} \rightarrow \mathbf{Z}^+$, denoted by $f(n) = n!$. The value of $f(n) = n!$ is the product of the first n positive integers, so $f(n) = 1 \cdot 2 \cdots (n - 1) \cdot n$ [and $f(0) = 0! = 1$].

EXAMPLE 33 We have $f(1) = 1! = 1$, $f(2) = 2! = 1 \cdot 2 = 2$, $f(6) = 6! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720$, and $f(20) = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 \cdot 11 \cdot 12 \cdot 13 \cdot 14 \cdot 15 \cdot 16 \cdot 17 \cdot 18 \cdot 19 \cdot 20 = 2,432,902,008,176,640,000$. 

Example 33 illustrates that the factorial function grows extremely rapidly as n grows. The rapid growth of the factorial function is made clearer by Stirling's formula, a result from higher mathematics that tells us that $n! \sim \sqrt{2\pi n}(n/e)^n$. Here, we have used the notation $f(n) \sim g(n)$, which means that the ratio $f(n)/g(n)$ approaches 1 as n grows without bound (that is, $\lim_{n \rightarrow \infty} f(n)/g(n) = 1$). The symbol $\sim$ is read "is asymptotic to." Stirling's formula is named after James Stirling, a Scottish mathematician of the eighteenth century.

JAMES STIRLING (1692–1770) James Stirling was born near the town of Stirling, Scotland. His family strongly supported the Jacobite cause of the Stuarts as an alternative to the British crown. The first information known about James is that he entered Balliol College, Oxford, on a scholarship in 1711. However, he later lost his scholarship when he refused to pledge his allegiance to the British crown. The first Jacobean rebellion took place in 1715, and Stirling was accused of communicating with rebels. He was charged with cursing King George, but he was acquitted of these charges. Even though he could not graduate from Oxford because of his politics, he remained there for several years. Stirling published his first work, which extended Newton's work on plane curves, in 1717. He traveled to Venice, where a chair of mathematics had been promised to him, an appointment that unfortunately fell through. Nevertheless, Stirling stayed in Venice, continuing his mathematical work. He attended the University of Padua in 1721, and in 1722 he returned to Glasgow. Stirling apparently fled Italy after learning the secrets of the Italian glass industry, avoiding the efforts of Italian glass makers to assassinate him to protect their secrets.

In late 1724 Stirling moved to London, staying there 10 years teaching mathematics and actively engaging in research. In 1730 he published *Methodus Differentialis*, his most important work, presenting results on infinite series, summations, interpolation, and quadrature. It is in this book that his asymptotic formula for $n!$ appears. Stirling also worked on gravitation and the shape of the earth; he stated, but did not prove, that the earth is an oblate spheroid. Stirling returned to Scotland in 1735, when he was appointed manager of a Scottish mining company. He was very successful in this role and even published a paper on the ventilation of mine shafts. He continued his mathematical research, but at a reduced pace, during his years in the mining industry. Stirling is also noted for surveying the River Clyde with the goal of creating a series of locks to make it navigable. In 1752 the citizens of Glasgow presented him with a silver teakettle as a reward for this work.

2.3.6 Partial Functions

A program designed to evaluate a function may not produce the correct value of the function for all elements in the domain of this function. For example, a program may not produce a correct value because evaluating the function may lead to an infinite loop or an overflow. Similarly, in abstract mathematics, we often want to discuss functions that are defined only for a subset of the real numbers, such as $1/x$, $\sqrt{x}$, and $\arcsin(x)$. Also, we may want to use such notions as the “youngest child” function, which is undefined for a couple having no children, or the “time of sunrise,” which is undefined for some days above the Arctic Circle. To study such situations, we use the concept of a partial function.

Definition 13

A *partial function* f from a set A to a set B is an assignment to each element a in a subset of A , called the *domain of definition* of f , of a unique element b in B . The sets A and B are called the *domain* and *codomain* of f , respectively. We say that f is *undefined* for elements in A that are not in the domain of definition of f . When the domain of definition of f equals A , we say that f is a *total function*.

Remark: We write $f : A \rightarrow B$ to denote that f is a partial function from A to B . Note that this is the same notation as is used for functions. The context in which the notation is used determines whether f is a partial function or a total function.

EXAMPLE 34

The function $f : \mathbf{Z} \rightarrow \mathbf{R}$ where $f(n) = \sqrt{n}$ is a partial function from $\mathbf{Z}$ to $\mathbf{R}$ where the domain of definition is the set of nonnegative integers. Note that f is undefined for negative integers. ◀

Exercises

- Why is f not a function from $\mathbf{R}$ to $\mathbf{R}$ if
 - $f(x) = 1/x$?
 - $f(x) = \sqrt{x}$?
 - $f(x) = \pm\sqrt{(x^2 + 1)}$?
- Determine whether f is a function from $\mathbf{Z}$ to $\mathbf{R}$ if
 - $f(n) = \pm n$.
 - $f(n) = \sqrt{n^2 + 1}$.
 - $f(n) = 1/(n^2 - 4)$.
- Determine whether f is a function from the set of all bit strings to the set of integers if
 - $f(S)$ is the position of a 0 bit in S .
 - $f(S)$ is the number of 1 bits in S .
 - $f(S)$ is the smallest integer i such that the i th bit of S is 1 and $f(S) = 0$ when S is the empty string, the string with no bits.
- Find the domain and range of these functions. Note that in each case, to find the domain, determine the set of elements assigned values by the function.
 - the function that assigns to each nonnegative integer its last digit
 - the function that assigns the next largest integer to a positive integer
 - the function that assigns to a bit string the number of one bits in the string
 - the function that assigns to a bit string the number of bits in the string
- Find the domain and range of these functions. Note that in each case, to find the domain, determine the set of elements assigned values by the function.
 - the function that assigns to each bit string the number of ones in the string minus the number of zeros in the string
 - the function that assigns to each bit string twice the number of zeros in that string
 - the function that assigns the number of bits left over when a bit string is split into bytes (which are blocks of 8 bits)
 - the function that assigns to each positive integer the largest perfect square not exceeding this integer
- Find the domain and range of these functions.
 - the function that assigns to each pair of positive integers the first integer of the pair
 - the function that assigns to each positive integer its largest decimal digit
 - the function that assigns to a bit string the number of ones minus the number of zeros in the string
 - the function that assigns to each positive integer the largest integer not exceeding the square root of the integer
 - the function that assigns to a bit string the longest string of ones in the string

7. Find the domain and range of these functions.
 - a) the function that assigns to each pair of positive integers the maximum of these two integers
 - b) the function that assigns to each positive integer the number of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 that do not appear as decimal digits of the integer
 - c) the function that assigns to a bit string the number of times the block 11 appears
 - d) the function that assigns to a bit string the numerical position of the first 1 in the string and that assigns the value 0 to a bit string consisting of all 0s
8. Find these values.

a) $\lfloor 1.1 \rfloor$	b) $\lceil 1.1 \rceil$
c) $\lfloor -0.1 \rfloor$	d) $\lceil -0.1 \rceil$
e) $\lfloor 2.99 \rfloor$	f) $\lceil -2.99 \rceil$
g) $\lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil$	h) $\lceil \lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil + \frac{1}{2} \rceil$
9. Find these values.

a) $\lceil \frac{3}{4} \rceil$	b) $\lfloor \frac{7}{8} \rfloor$
c) $\lceil -\frac{3}{4} \rceil$	d) $\lfloor -\frac{7}{8} \rfloor$
e) $\lceil 3 \rceil$	f) $\lfloor -1 \rfloor$
g) $\lfloor \frac{1}{2} + \lceil \frac{3}{2} \rceil \rfloor$	h) $\lfloor \frac{1}{2} \cdot \lfloor \frac{5}{2} \rfloor \rfloor$
10. Determine whether each of these functions from $\{a, b, c, d\}$ to itself is one-to-one.
 - a) $f(a) = b, f(b) = a, f(c) = c, f(d) = d$
 - b) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$
 - c) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$
11. Which functions in Exercise 10 are onto?
12. Determine whether each of these functions from $\mathbf{Z}$ to $\mathbf{Z}$ is one-to-one.

a) $f(n) = n - 1$	b) $f(n) = n^2 + 1$
c) $f(n) = n^3$	d) $f(n) = \lfloor n/2 \rfloor$
13. Which functions in Exercise 12 are onto?
14. Determine whether $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}$ is onto if
 - a) $f(m, n) = 2m - n$.
 - b) $f(m, n) = m^2 - n^2$.
 - c) $f(m, n) = m + n + 1$.
 - d) $f(m, n) = |m| - |n|$.
 - e) $f(m, n) = m^2 - 4$.
15. Determine whether the function $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}$ is onto if
 - a) $f(m, n) = m + n$.
 - b) $f(m, n) = m^2 + n^2$.
 - c) $f(m, n) = m$.
 - d) $f(m, n) = |n|$.
 - e) $f(m, n) = m - n$.
16. Consider these functions from the set of students in a discrete mathematics class. Under what conditions is the function one-to-one if it assigns to a student his or her
 - a) mobile phone number.
 - b) student identification number.
 - c) final grade in the class.
 - d) home town.
17. Consider these functions from the set of teachers in a school. Under what conditions is the function one-to-one if it assigns to a teacher his or her
 - a) office.
 - b) assigned bus to chaperone in a group of buses taking students on a field trip.
 - c) salary.
 - d) social security number.
18. Specify a codomain for each of the functions in Exercise 16. Under what conditions is each of these functions with the codomain you specified onto?
19. Specify a codomain for each of the functions in Exercise 17. Under what conditions is each of the functions with the codomain you specified onto?
20. Give an example of a function from $\mathbf{N}$ to $\mathbf{N}$ that is
 - a) one-to-one but not onto.
 - b) onto but not one-to-one.
 - c) both onto and one-to-one (but different from the identity function).
 - d) neither one-to-one nor onto.
21. Give an explicit formula for a function from the set of integers to the set of positive integers that is
 - a) one-to-one, but not onto.
 - b) onto, but not one-to-one.
 - c) one-to-one and onto.
 - d) neither one-to-one nor onto.
22. Determine whether each of these functions is a bijection from $\mathbf{R}$ to $\mathbf{R}$.
 - a) $f(x) = -3x + 4$
 - b) $f(x) = -3x^2 + 7$
 - c) $f(x) = (x + 1)/(x + 2)$
 - d) $f(x) = x^5 + 1$
23. Determine whether each of these functions is a bijection from $\mathbf{R}$ to $\mathbf{R}$.
 - a) $f(x) = 2x + 1$
 - b) $f(x) = x^2 + 1$
 - c) $f(x) = x^3$
 - d) $f(x) = (x^2 + 1)/(x^2 + 2)$
24. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ and let $f(x) > 0$ for all $x \in \mathbf{R}$. Show that $f(x)$ is strictly increasing if and only if the function $g(x) = 1/f(x)$ is strictly decreasing.
25. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ and let $f(x) > 0$ for all $x \in \mathbf{R}$. Show that $f(x)$ is strictly decreasing if and only if the function $g(x) = 1/f(x)$ is strictly increasing.
26. a) Prove that a strictly increasing function from $\mathbf{R}$ to itself is one-to-one.
 b) Give an example of an increasing function from $\mathbf{R}$ to itself that is not one-to-one.
27. a) Prove that a strictly decreasing function from $\mathbf{R}$ to itself is one-to-one.
 b) Give an example of a decreasing function from $\mathbf{R}$ to itself that is not one-to-one.
28. Show that the function $f(x) = e^x$ from the set of real numbers to the set of real numbers is not invertible, but if the codomain is restricted to the set of positive real numbers, the resulting function is invertible.

29. Show that the function $f(x) = |x|$ from the set of real numbers to the set of nonnegative real numbers is not invertible, but if the domain is restricted to the set of nonnegative real numbers, the resulting function is invertible.
30. Let $S = \{-1, 0, 2, 4, 7\}$. Find $f(S)$ if
- $f(x) = 1$.
 - $f(x) = 2x + 1$.
 - $f(x) = \lfloor x/5 \rfloor$.
 - $f(x) = \lfloor (x^2 + 1)/3 \rfloor$.
31. Let $f(x) = \lfloor x^2/3 \rfloor$. Find $f(S)$ if
- $S = \{-2, -1, 0, 1, 2, 3\}$.
 - $S = \{0, 1, 2, 3, 4, 5\}$.
 - $S = \{1, 5, 7, 11\}$.
 - $S = \{2, 6, 10, 14\}$.
32. Let $f(x) = 2x$ where the domain is the set of real numbers. What is
- $f(\mathbf{Z})$?
 - $f(\mathbf{N})$?
 - $f(\mathbf{R})$?
33. Suppose that g is a function from A to B and f is a function from B to C .
- Show that if both f and g are one-to-one functions, then $f \circ g$ is also one-to-one.
 - Show that if both f and g are onto functions, then $f \circ g$ is also onto.
34. Suppose that g is a function from A to B and f is a function from B to C . Prove each of these statements.
- If $f \circ g$ is onto, then f must also be onto.
 - If $f \circ g$ is one-to-one, then g must also be one-to-one.
 - If $f \circ g$ is a bijection, then g is onto if and only if f is one-to-one.
35. Find an example of functions f and g such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.
- *36. If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? Justify your answer.
- *37. If f and $f \circ g$ are onto, does it follow that g is onto? Justify your answer.
38. Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 2$, are functions from $\mathbf{R}$ to $\mathbf{R}$.
39. Find $f + g$ and fg for the functions f and g given in Exercise 36.
40. Let $f(x) = ax + b$ and $g(x) = cx + d$, where a, b, c , and d are constants. Determine necessary and sufficient conditions on the constants a, b, c , and d so that $f \circ g = g \circ f$.
41. Show that the function $f(x) = ax + b$ from $\mathbf{R}$ to $\mathbf{R}$, where a and b are constants with $a \neq 0$ is invertible, and find the inverse of f .
42. Let f be a function from the set A to the set B . Let S and T be subsets of A . Show that
- $f(S \cup T) = f(S) \cup f(T)$.
 - $f(S \cap T) \subseteq f(S) \cap f(T)$.
43. a) Give an example to show that the inclusion in part (b) in Exercise 42 may be proper.
- b) Show that if f is one-to-one, the inclusion in part (b) in Exercise 42 is an equality.
- Let f be a function from the set A to the set B . Let S be a subset of B . We define the **inverse image** of S to be the subset of A whose elements are precisely all preimages of all elements of S . We denote the inverse image of S by $f^{-1}(S)$, so $f^{-1}(S) = \{a \in A \mid f(a) \in S\}$. [Beware: The notation f^{-1} is used in two different ways. Do not confuse the notation introduced here with the notation $f^{-1}(y)$ for the value at y of the inverse of the invertible function f . Notice also that $f^{-1}(S)$, the inverse image of the set S , makes sense for all functions f , not just invertible functions.]
44. Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ defined by $f(x) = x^2$. Find
- $f^{-1}(\{1\})$.
 - $f^{-1}(\{x \mid 0 < x < 1\})$.
 - $f^{-1}(\{x \mid x > 4\})$.
45. Let $g(x) = \lfloor x \rfloor$. Find
- $g^{-1}(\{0\})$.
 - $g^{-1}(\{-1, 0, 1\})$.
 - $g^{-1}(\{x \mid 0 < x < 1\})$.
46. Let f be a function from A to B . Let S and T be subsets of B . Show that
- $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.
 - $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.
47. Let f be a function from A to B . Let S be a subset of B . Show that $f^{-1}(\bar{S}) = \overline{f^{-1}(S)}$.
48. Show that $\lfloor x + \frac{1}{2} \rfloor$ is the closest integer to the number x , except when x is midway between two integers, when it is the larger of these two integers.
49. Show that $\lfloor x - \frac{1}{2} \rfloor$ is the closest integer to the number x , except when x is midway between two integers, when it is the smaller of these two integers.
50. Show that if x is a real number, then $\lceil x \rceil - \lfloor x \rfloor = 1$ if x is not an integer and $\lceil x \rceil - \lfloor x \rfloor = 0$ if x is an integer.
51. Show that if x is a real number, then $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$.
52. Show that if x is a real number and m is an integer, then $\lceil x + m \rceil = \lceil x \rceil + m$.
53. Show that if x is a real number and n is an integer, then
- $x < n$ if and only if $\lfloor x \rfloor < n$.
 - $n < x$ if and only if $n < \lceil x \rceil$.
54. Show that if x is a real number and n is an integer, then
- $x \leq n$ if and only if $\lfloor x \rfloor \leq n$.
 - $n \leq x$ if and only if $n \leq \lceil x \rceil$.
55. Prove that if n is an integer, then $\lfloor n/2 \rfloor = n/2$ if n is even and $(n - 1)/2$ if n is odd.
56. Prove that if x is a real number, then $\lfloor -x \rfloor = -\lceil x \rceil$ and $\lceil -x \rceil = -\lfloor x \rfloor$.
57. The function INT is found on some calculators, where $\text{INT}(x) = \lfloor x \rfloor$ when x is a nonnegative real number and $\text{INT}(x) = \lceil x \rceil$ when x is a negative real number. Show that this INT function satisfies the identity $\text{INT}(-x) = -\text{INT}(x)$.

58. Let a and b be real numbers with $a < b$. Use the floor and/or ceiling functions to express the number of integers n that satisfy the inequality $a \leq n \leq b$.
59. Let a and b be real numbers with $a < b$. Use the floor and/or ceiling functions to express the number of integers n that satisfy the inequality $a < n < b$.
60. How many bytes are required to encode n bits of data where n equals
 a) 4? b) 10? c) 500? d) 3000?
61. How many bytes are required to encode n bits of data where n equals
 a) 7? b) 17? c) 1001? d) 28,800?
62. How many ATM cells (described in Example 30) can be transmitted in 10 seconds over a link operating at the following rates?
 a) 128 kilobits per second (1 kilobit = 1000 bits)
 b) 300 kilobits per second
 c) 1 megabit per second (1 megabit = 1,000,000 bits)
63. Data are transmitted over a particular Ethernet network in blocks of 1500 octets (blocks of 8 bits). How many blocks are required to transmit the following amounts of data over this Ethernet network? (Note that a byte is a synonym for an octet, a kilobyte is 1000 bytes, and a megabyte is 1,000,000 bytes.)
 a) 150 kilobytes of data
 b) 384 kilobytes of data
 c) 1.544 megabytes of data
 d) 45.3 megabytes of data
64. Draw the graph of the function $f(n) = 1 - n^2$ from $\mathbf{Z}$ to $\mathbf{Z}$.
65. Draw the graph of the function $f(x) = \lfloor 2x \rfloor$ from $\mathbf{R}$ to $\mathbf{R}$.
66. Draw the graph of the function $f(x) = \lfloor x/2 \rfloor$ from $\mathbf{R}$ to $\mathbf{R}$.
67. Draw the graph of the function $f(x) = \lfloor x \rfloor + \lfloor x/2 \rfloor$ from $\mathbf{R}$ to $\mathbf{R}$.
68. Draw the graph of the function $f(x) = \lfloor x \rfloor + \lfloor x/2 \rfloor$ from $\mathbf{R}$ to $\mathbf{R}$.
69. Draw graphs of each of these functions.
 a) $f(x) = \lfloor x + \frac{1}{2} \rfloor$ b) $f(x) = \lfloor 2x + 1 \rfloor$
 c) $f(x) = \lfloor x/3 \rfloor$ d) $f(x) = \lceil 1/x \rceil$
 e) $f(x) = \lfloor x - 2 \rfloor + \lfloor x + 2 \rfloor$
 f) $f(x) = \lfloor 2x \rfloor \lfloor x/2 \rfloor$ g) $f(x) = \lceil \lfloor x - \frac{1}{2} \rfloor + \frac{1}{2} \rceil$
70. Draw graphs of each of these functions.
 a) $f(x) = \lfloor 3x - 2 \rfloor$ b) $f(x) = \lfloor 0.2x \rfloor$
 c) $f(x) = \lfloor -1/x \rfloor$ d) $f(x) = \lfloor x^2 \rfloor$
 e) $f(x) = \lfloor x/2 \rfloor \lfloor x/2 \rfloor$ f) $f(x) = \lfloor x/2 \rfloor + \lfloor x/2 \rfloor$
 g) $f(x) = \lfloor 2 \lfloor x/2 \rfloor + \frac{1}{2} \rfloor$
71. Find the inverse function of $f(x) = x^3 + 1$.
72. Suppose that f is an invertible function from Y to Z and g is an invertible function from X to Y . Show that the inverse of the composition $f \circ g$ is given by $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.
73. Let S be a subset of a universal set U . The **characteristic function** f_S of S is the function from U to the set $\{0, 1\}$ such that $f_S(x) = 1$ if x belongs to S and $f_S(x) = 0$ if x does not belong to S . Let A and B be sets. Show that for all $x \in U$,
 a) $f_{A \cap B}(x) = f_A(x) \cdot f_B(x)$
 b) $f_{A \cup B}(x) = f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)$
 c) $f_{\bar{A}}(x) = 1 - f_A(x)$
 d) $f_{A \oplus B}(x) = f_A(x) + f_B(x) - 2f_A(x)f_B(x)$
74. Suppose that f is a function from A to B , where A and B are finite sets with $|A| = |B|$. Show that f is one-to-one if and only if it is onto.
75. Prove or disprove each of these statements about the floor and ceiling functions.
 a) $\lceil \lfloor x \rfloor \rceil = \lfloor x \rfloor$ for all real numbers x .
 b) $\lfloor 2x \rfloor = 2 \lfloor x \rfloor$ whenever x is a real number.
 c) $\lfloor x \rfloor + \lfloor y \rfloor - \lfloor x + y \rfloor = 0$ or 1 whenever x and y are real numbers.
 d) $\lfloor xy \rfloor = \lfloor x \rfloor \lfloor y \rfloor$ for all real numbers x and y .
 e) $\left\lfloor \frac{x}{2} \right\rfloor = \left\lfloor \frac{x+1}{2} \right\rfloor$ for all real numbers x .
76. Prove or disprove each of these statements about the floor and ceiling functions.
 a) $\lfloor \lceil x \rceil \rfloor = \lceil x \rceil$ for all real numbers x .
 b) $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$ for all real numbers x and y .
 c) $\lceil \lfloor x/2 \rfloor / 2 \rceil = \lceil x/4 \rceil$ for all real numbers x .
 d) $\lfloor \sqrt{\lfloor x \rfloor} \rfloor = \lfloor \sqrt{x} \rfloor$ for all positive real numbers x .
 e) $\lfloor x \rfloor + \lfloor y \rfloor + \lfloor x + y \rfloor \leq \lfloor 2x \rfloor + \lfloor 2y \rfloor$ for all real numbers x and y .
77. Prove that if x is a positive real number, then
 a) $\lfloor \sqrt{\lfloor x \rfloor} \rfloor = \lfloor \sqrt{x} \rfloor$.
 b) $\lceil \sqrt{\lceil x \rceil} \rceil = \lceil \sqrt{x} \rceil$.
78. Let x be a real number. Show that $\lfloor 3x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{3} \rfloor + \lfloor x + \frac{2}{3} \rfloor$.
79. For each of these partial functions, determine its domain, codomain, domain of definition, and the set of values for which it is undefined. Also, determine whether it is a total function.
 a) $f: \mathbf{Z} \rightarrow \mathbf{R}, f(n) = 1/n$
 b) $f: \mathbf{Z} \rightarrow \mathbf{Z}, f(n) = \lfloor n/2 \rfloor$
 c) $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Q}, f(m, n) = m/n$
 d) $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}, f(m, n) = mn$
 e) $f: \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}, f(m, n) = m - n$ if $m > n$
80. a) Show that a partial function from A to B can be viewed as a function f^* from A to $B \cup \{u\}$, where u is not an element of B and

$$f^*(a) = \begin{cases} f(a) & \text{if } a \text{ belongs to the domain} \\ & \text{of definition of } f \\ u & \text{if } f \text{ is undefined at } a. \end{cases}$$
 b) Using the construction in (a), find the function f^* corresponding to each partial function in Exercise 79.
81. a) Show that if a set S has cardinality m , where m is a positive integer, then there is a one-to-one correspondence between S and the set $\{1, 2, \dots, m\}$.
 b) Show that if S and T are two sets each with m elements, where m is a positive integer, then there is a one-to-one correspondence between S and T .
- *82. Show that a set S is infinite if and only if there is a proper subset A of S such that there is a one-to-one correspondence between A and S .